WEAVE: Affordable Real Style Transfer via Wavelet-Endpoint Aligned Vector-Field Editing

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Abstract

Current style transfer faces two main issues. First, state-of-the-art diffusion models , along with training-free methods built on them, require more than 60 GB of VRAM therfore cannot run on consumer GPUs. Second, lightweight methods without these generative backbones often underperform a trivial identity mapping, which scores high on traditional metrics like ArtFID and LPIPS but was rarely evaluated as a baseline. Therefore, an effective style transfer method should outperform the identity baseline while remaining computationally affordable.

This tendency to underperform identity may stem from how natural images contain much more low-frequency energy than high-frequency details. When a single model learns all frequencies together in a shared space, it tends to prioritize fitting the strong low-frequency signals to minimize the training loss. Consequently, the learning of high-frequency textures is suppressed, making the model more likely to converge toward identity-like solutions or fail to capture target stylistic patterns.

To address this, we propose WEAVE, using wavelet decomposition to separate low-frequency content from high-frequency style details. We operate in the VAE latent space for computational efficiency, preserving style textures rather than high-frequency pixel-level noise. We further replace point-wise MSE with latent Sliced Wasserstein Distance (SWD) to reduce style entanglement, while routing style supervision exclusively through high-frequency subbands.

We also introduce an identity-aware evaluation protocol that detects when conventional metrics favor the identity mapping. Under this protocol, WEAVE clears the identity baseline. With only 903K parameters, it trains in 3 minutes on a single consumer GPU, versus hours for billion-parameter editors, making high-quality style transfer both effective and affordable.

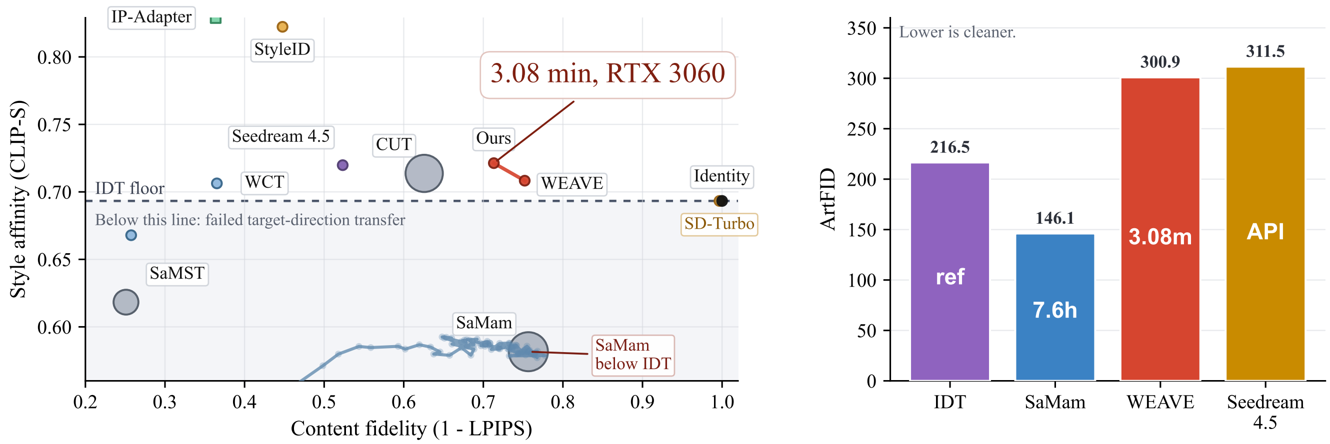


Figure 1: IDT-calibrated comparison on Distinct5-WikiArt. Left: CLIP-S versus $1 - \text{LPIPS}$, with bubble area encoding training time. The red points mark two WEAVE operating points. Square markers denote training-free diffusion editors (StyleAligned, IP-Adapter). The dashed line marks the IDT floor (0.6933); methods below it have not moved in the requested direction. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

Introduction

Style transfer faces two problems in deployment and evaluation. On one hand, state-of-the-art diffusion models (e.g., Qwen-Image and FLUX.2) and their training-free variants demand more than 60 GB of GPU memory, making them unusable on consumer hardware. On the other hand, lightweight alternatives that avoid these heavy generative backbones often struggle to achieve genuine style transfer. Specifically, the style affinity of outputs from many lightweight models often falls below that of the unchanged source image (i.e., the identity mapping). This problem has been hidden by conventional evaluation protocols, such as LPIPS and ArtFID, which heavily reward structural reconstruction but rarely include the identity mapping as an explicit baseline. As visualized in Figure 1 (Left), when the identity mapping is explicitly plotted, modern lightweight methods like SaMam clearly fall below this natural baseline. To fix this, we introduce the Identical-Image Transfer (IDT) calibration, transforming standard metrics into direction-aware measures.

Why do modern lightweight architectures, despite their modern design, consistently fall into this identity trap? We argue that the root cause is not a lack of model capacity, but a conflict in the latent space. In typical flow-matching frameworks, the model predicts a velocity field that transports the source latent toward the target. However, natural images follow a $1/f$ spectral distribution, meaning the low-frequency structural energy (e.g., global layouts and object shapes) is orders of magnitude stronger than the high-frequency stylistic textures (e.g., brushstrokes and material patterns). When minimizing reconstruction loss in a shared latent space, the optimizer naturally prioritizes fitting the massive low-frequency errors. Consequently, the subtle style-conditioned gradients are overwhelmed, pushing the model toward a style-agnostic identity mapping. This is why simply scaling up parameters fails: structure and style compete in the same coordinate basis.

To solve the conflict between structure and style, we propose WEAVE. First, we must choose where to learn this transfer. Training a small model directly on raw pixels is slow and struggles with pixel noise, while full diffusion models provide clean features but demand massive GPU memory. We choose a practical middle ground: we use the pre-trained VAE from diffusion models to compress images into a clean, low-dimensional space, but we completely discard the heavy diffusion network itself. Instead, we build a tiny network (only 903K parameters) to learn the style transfer within this clean space.

Within this space, we apply a wavelet transform to separate the features into structure (low frequency) and texture (high frequency), which prevents the strong structure from hiding the subtle style. However, matching the overall texture often causes obvious errors, such as blending sky patterns onto human skin. To prevent this, we use the low-frequency structure to group similar object regions and match textures only within the same region (via Sliced Wasserstein Distance). Finally, to prevent content from degrading across generation steps, we apply this texture matching only at the very last step. Practical deployment demands not only efficient inference but also affordable retraining when adapting to new artistic domains. This streamlined design allows WEAVE to clear the identity baseline and finish training in just 3 minutes on a single consumer GPU. Because our alignment mechanism operates directly on the latent features, it can also be applied as a training-free approach to large pre-trained models.

Contributions. (1) **Evaluation Protocol:** We introduce the Identical-Image Transfer (IDT) calibration, exposing the blind spot where lightweight models underperform the unchanged source. (2) **Representation & Methodology:** We reveal the frequency-dominance failure mode in shared latent spaces and propose WEAVE, a wavelet-decoupled framework that achieves genuine style transfer with only 903K parameters. (3) **Efficiency & Generalization:** WEAVE trains in just 3 minutes on a single consumer GPU, and its latent operations serve as a model-agnostic, training-free plugin for large diffusion backbones.

Related Work

Evolution of Style Transfer

Early style transfer relied on optimizing pixel spaces (Gatys, Ecker, and Bethge 2015) or matching feature statistics in pre-trained discriminative networks, such as AdaIN (Huang and Belongie 2017) and WCT (Li et al. 2017). Subsequent works introduced GANs and contrastive learning for unpaired translation, notably CUT (Park et al. 2020). Diffusion models moved style transfer from explicit feature transformation to generative editing in latent spaces. Recently, the field has been dominated by heavy generative priors based on latent diffusion models (Rombach et al. 2022). State-of-the-art backbones like FLUX.2 and Seedream, along with training-free attention manipulation techniques like StyleID (Chung, Hyun, and Heo 2024) and StyleAligned (Hertz et al. 2024), achieve strong style transfer performance by leveraging massive pre-trained representations.

Lightweight Architectures and Evaluation Protocols

To reduce computational costs, modern lightweight architectures, such as Mamba-based SaMST and SaMam (Liu et al. 2024, 2025), have been explored. While these methods improve efficiency, standard evaluation protocols (Zhang et al. 2018; Wright and Ommer 2022) heavily reward source reconstruction and historically omit the identity mapping as an explicit baseline. These observations highlight the importance of explicitly considering identity mappings when evaluating style transfer.

Spectral Bias and Frequency-aware Optimization

Neural networks exhibit a well-documented spectral bias (Rahaman et al. 2019), preferentially fitting low-frequency signals. Combined with the $1/f$ spectral distribution of natural images, this observation motivates frequency-aware optimization strategies.

Wavelet and Latent-space Alignment

Wavelet transforms have been widely adopted in image restoration and generative modeling to separate multi-scale information, and have been adapted for style transfer (Yoo et al. 2019) and diffusion models (Phung, Dao, and Tran 2023). Concurrently, latent diffusion models (Rombach et al. 2022) introduced perceptual latent spaces that compress high-dimensional pixels into regularized representations. This regularized distribution has facilitated statistical alignment methods such as whitening-and-coloring transforms (Li et al. 2017) and sliced Wasserstein distance, making distribution matching more stable in latent spaces.

Method

We describe WEAVE following the logic of the Introduction: first, where to learn the transfer (VAE latent space); second, how to separate conflicting signals (wavelet decomposition); third, how to route style information while protecting content (frequency-aware architecture and endpoint alignment); and fourth, how the decoupled design enables few-shot adaptation.

Framework Setup

Let $x_c \in \mathbb{R}^{H\times W\times 3}$ be a content image and $s \in \{1,\dots,K\}$ a target style domain. We operate in the latent space of the Stable Diffusion v1.5 EMA VAE (Rombach et al. 2022). We write $z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4\times 32\times 32}$. During training, $z_1 \sim p_s$ is a latent sampled from the target domain. Rectified flow uses the straight segment

$$z_t = (1-t)z_0 + tz_1, \quad u_t = z_1 - z_0, \quad t \sim \mathcal{U}([0,1]). \tag{1}$$

At inference we integrate from z_0 to $\hat{z}_1$ with an 8-step Heun solver and decode $\hat{x}_s = \mathcal{D}(\hat{z}_1)$.

IDT Calibration. Style transfer can suffer from a no-op shortcut: when the source already contains stylistic features, standard metrics can reward the identity mapping. We formalize this through **Identical-Image Transfer (IDT)**, where the unchanged source defines a floor that any valid method must clear.

Definition 1 (Identical-Image Transfer (IDT) Floor)**.** *Because the source image is already stylized,* $\mathrm{CLIP}(x_{\mathrm{src}}, s_{\mathrm{tgt}})$ *is a natural no-op baseline. A valid transfer must satisfy* $\mathrm{CLIP}(x_{\mathrm{out}}, s_{\mathrm{tgt}}) \geq \mathrm{CLIP}(x_{\mathrm{src}}, s_{\mathrm{tgt}})$*; below-floor scores indicate negative transfer—the method has not moved in the requested direction.*

The IDT floor turns standard metrics into direction-aware measures: low LPIPS alone is not enough—a method must also clear the floor, or it is merely reconstructing the source. On Distinct5-WikiArt, SaMam achieves LPIPS 0.2434 but its CLIP-S of 0.5816 falls below the IDT floor of 0.6933—a clear signature of source reconstruction rather than target-direction transfer.

Wavelet Decomposition and Frequency Routing

As discussed in Section 2.3, natural images follow a $1/f$ spectral distribution: low-frequency structural energy dominates over high-frequency textures. Learning style transfer in a shared latent space makes the optimizer prioritize structure over style. WEAVE addresses this by changing coordinates.

Haar wavelet decomposition. A single-level Haar wavelet transform decomposes the $4\times 32\times 32$ latent into four orthogonal subbands: a low-frequency LL band and three high-frequency bands (LH, HL, HH), each of size $4\times 16\times 16$. We write $\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and $\mathcal{W}(u_t) = (u_\ell, u_1, u_2, u_3)$. The transform is orthogonal, so the Frobenius norm splits additively:

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2. \tag{2}$$

Due to the orthogonality of the Haar transform, modifying only the high-frequency subbands leaves the LL coordinate unchanged—structure is preserved by construction. Haar is selected for its exact orthogonality, compact spatial support, and very low cost; a single decomposition level suffices for the compact latent resolution.

Empirical frequency probes. Across 5,000 random cross-style latent pairs, LL contains 69.5% of the squared target-transport energy, while normalized style separation is strongest in HH (Figure 3). These observations motivate weak LL supervision and greater emphasis on high-frequency appearance statistics.

Architecture and high-frequency routing. The velocity network consists of four residual blocks (width 64, $\sim$903K trainable parameters) with three separate prediction heads—one 1×1 convolution per supervised subband (ℓ, h_1, h_2). The h_3 (HH) band receives no velocity head and evolves passively along the rectified-flow trajectory; it is stylized only at the endpoint (Section). Style conditioning is injected through a memory module: each target style s owns a learned embedding $m_s \in \mathbb{R}^d$ (d=64), which is projected to keys and values for cross-attention. The attention query is assembled from the high-frequency subbands via a lightweight linear projection $Q[h_1; h_2; h_3]$, and the resulting features are fused into the residual-stream activations. Critically, the LL subband enters the backbone as a separate channel and interacts with style only through the shared residual blocks—never through the attention path—preserving the intended frequency separation. We use RMSNorm throughout, as GroupNorm erases the channel-wise mean carrying tonal information.

The overall architecture is shown in Figure 2.

The spectral flow-matching term is

$$\mathcal{L}_{\mathrm{impl}} := \mathbb{E}_t\left[\lambda_{LL}\|v_\ell - u_\ell\|_2^2 + \|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2\right], \tag{3}$$

with $\lambda_{LL} = 0.3$. WEAVE de-weights LL rather than eliminating it: some styles also shift coarse color tone, so LL remains a weak anchor. The predicted one-step endpoint is $\hat{z}_1 = z_0 + \mathcal{W}^{-1}(v_\ell, v_1, v_2, 0)$. At inference, style queries are routed exclusively through the high-frequency channels: $q_{\mathrm{routed}} = Q[h_{1,t}; h_{2,t}; h_{3,t}]$, $k = K(m_s)$, $v = V(m_s)$. During training, we use the high-frequency query with probability $p = 0.8$ and the full-latent query with probability 0.2, preserving a small low-frequency color channel while matching inference.

Endpoint Statistical Alignment

Why endpoint-only. Rectified flow interpolates latents along straight segments, but linearly interpolating high-frequency wavelet coefficients between two images causes variance decay: the variance of two independent textures drops parabolically to a minimum at $t = 0.5$, yielding blurred, phase-misaligned results. Per-step feature blending geometrically erodes the original signal: after n steps of blend strength α, only a fraction $(1-\alpha)^n$ of the content survives (e.g., dropping to 3.9×10^{-3} for $n = 8$, $\alpha = 0.5$). WEAVE therefore applies statistical alignment only once, at the final endpoint ($n = 1$), preserving structure regardless of style strength.

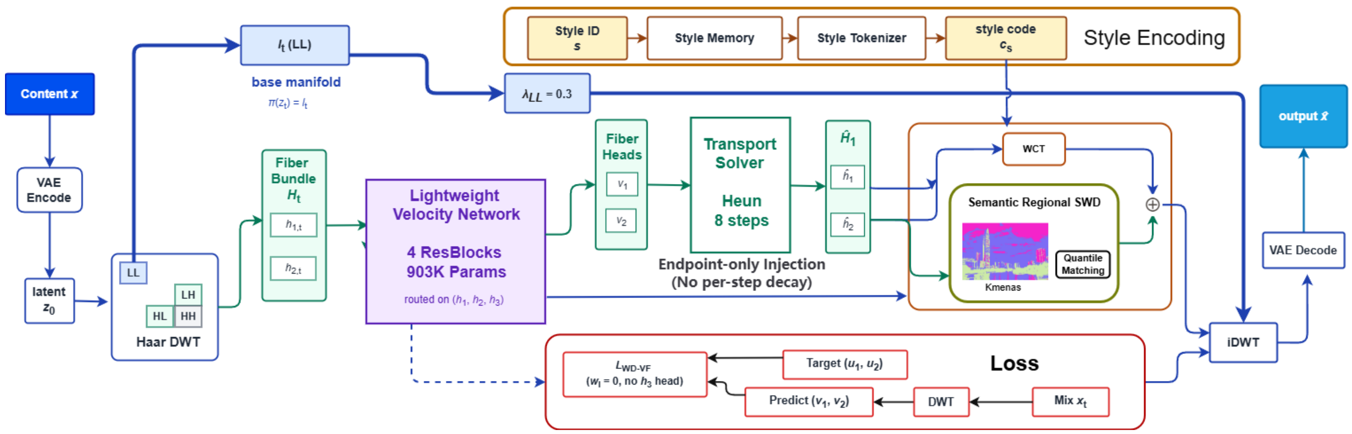


Figure 2: **WEAVE architecture**. A single-level Haar transform exposes LL/LH/HL/HH coordinates. The compact velocity network predicts LL/LH/HL motion, while style conditioning enters the shared backbone. Training combines band-weighted flow matching with an endpoint SWD guide.

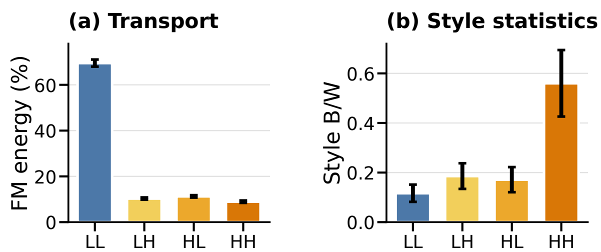


Figure 3: **Frequency probes.** (a) Cross-style transport energy in each Haar band. (b) Between-style versus within-style separation of latent statistics. Error bars summarize variation across style routes or statistics.

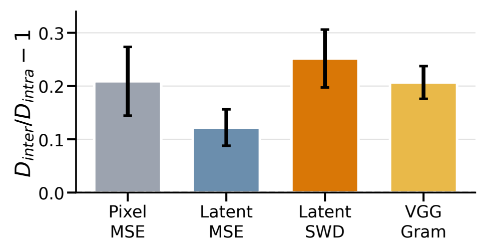


Figure 4: **Loss-level style separation.** Class-distance ratio on 150 Distinct5 images; zero means equal inter- and intra-style distance. Error bars are standard errors over ten style pairs.

Endpoint WCT. Let $\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3)$ and $\mathcal{W}(z_1^\star) = (\ell^\star, h_1^\star, h_2^\star, h_3^\star)$. For $i \in \{1, 2, 3\}$, the whitening-coloring map is

$$T_i(a) = \Sigma_i^{\star 1/2}\, \hat{\Sigma}_i^{-1/2}(a - \hat{\mu}_i) + \mu_i^\star, \qquad (4)$$

where $(\hat{\mu}_i, \hat{\Sigma}_i)$ and $(\mu_i^\star, \Sigma_i^\star)$ are the mean and covariance of $\hat{h}_i$ and $h_i^\star$. We set $\mathcal{W}(\hat{z}_1^+) = (\hat{\ell}, T_1(\hat{h}_1), T_2(\hat{h}_2), T_3(\hat{h}_3))$ with endpoint scales $(0.0, 0.3, 0.3, 0.5)$. Each T_i operates only on its own subband, so $\hat{\ell}$ is never touched—global structure is preserved. The VAE's KL regularization makes the latent approximately Gaussian (KL < 0.05 nats per subband), justifying second-order moment matching.

Sliced Wasserstein guide. Point-wise flow matching does not explicitly match the spatial feature distribution of the predicted endpoint. We therefore compare $\hat{z}_1$ and the sampled target latent z_1 with a global SWD:

$$\mathcal{L}_{\text{SWD}} = \frac{1}{64} \sum_{j=1}^{64} \left\| \text{sort}(\hat{Z} d_j) - \text{sort}(Z_1 d_j) \right\|_1. \qquad (5)$$

Here $\hat{Z}, Z_1 \in \mathbb{R}^{HW \times C}$ flatten the spatial dimensions; the 64 channel directions d_j are resampled on every call. In the current baseline this loss is global: semantic-region blending is disabled, and neither attention nor DWT-energy weighting is active. Thus SWD adds a distribution-level style signal that complements the paired spectral flow-matching target.

The probe in Figure 4 confirms this: point-wise MSE loses style separation after EMA-VAE encoding, whereas latent SWD restores a class-distance margin comparable to pixel-space and VGG-statistic losses.

Complete objective. The full training loss is

$$\mathcal{L} = \mathcal{L}_{\text{impl}} + 8\mathcal{L}_{\text{SWD}} + 0.1\mathcal{L}_{\text{edge}} + \mathcal{L}_{\text{low}}, \qquad (6)$$

where $\mathcal{L}_{\text{edge}}$ is an ℓ_1 match between high-pass residuals and $\mathcal{L}_{\text{low}}$ is an MSE anchor between low-pass endpoints. Ablation results (Section) confirm each component's contribution.

Few-Shot Adaptation

The high-frequency routing ($q_{\text{routed}} = Q[h_1; h_2; h_3]$) means the style query only sees the subbands (h_1, h_2, h_3), which together with the style-memory embedding m_s form a **style**

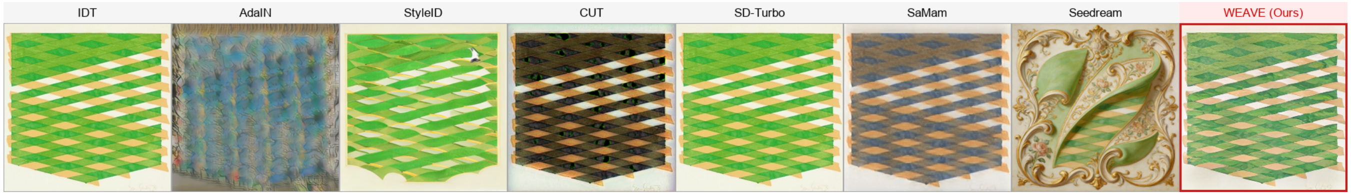


Figure 5: Extreme cross-domain stylization (Minimalism → Rococo). Heavy diffusion priors (Seedream, StyleID) hallucinate content, while classical methods and SaMam collapse to near-identity. WEAVE preserves source topology via low-frequency anchoring and injects the target's painterly textures exclusively through high-frequency subbands.

subspace that is approximately decoupled from the content (LL) subspace. The flow backbone thus learns a transport operator that is style-agnostic: s only enters through the high-frequency query path. When a new style is introduced, the transport dynamics stay valid; the only unknown is the target distribution of the high-frequency subbands, so few-shot adaptation reduces to estimating their mean and covariance from a few references—a low-dimensional problem, which is why 30 references suffice. Updating m_s does not interfere with content-preserving transport because the LL path is never touched by the style query.

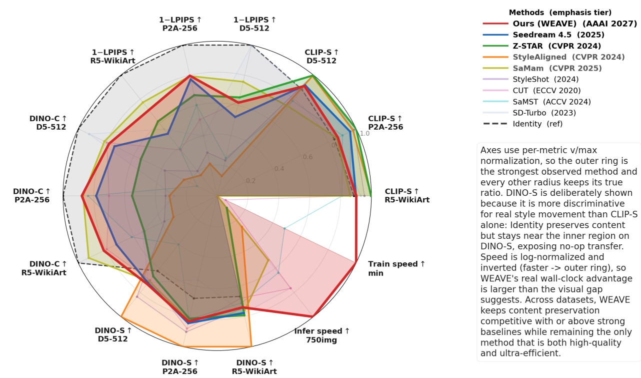
Adaptation protocol. For a new style with K reference images, we extract their latents, decompose them via Haar, and compute the target subband statistics $\{(\mu_i^\star, \Sigma_i^\star)\}_{i=1}^3$ for endpoint WCT. The style-memory embedding m_s is initialized as the mean-pooled CLIP image feature of the references and is then optimized—together with the backbone parameters—on the standard WEAVE objective for 200 steps (approximately 20 seconds on an RTX 3060). We evaluate two modes: *training-free* (freeze the backbone, update only m_s and the endpoint statistics from 30 references) and *fine-tuning* (update the full model). The training-free mode already clears the IDT floor on all eight held-out styles; fine-tuning adds only a modest $+0.008$ CLIP-S at substantially higher compute cost, confirming that the decoupled geometry makes the content-to-style mapping largely reusable across unseen domains.

Experiments

Setup and Protocol

Setup. We use Distinct5-WikiArt-512 with five styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) selected for having the lowest IDT CLIP-S, making accidental no-op success harder. Each domain has 3,600 training images and 150 test images; evaluation uses 750 source-target pairs, including 150 identity pairs for IDT calibration. For generalization, WEAVE is trained from scratch on all 20 WikiArt families and evaluated on all $20\times20\times30{=}12{,}000$ source-target pairs; baselines are evaluated on the same Distinct5 subset (750 pairs) to keep compute bounded. CLIP-S (ViT-B/32) and LPIPS (AlexNet) are the primary metrics. Baselines include Identity, AdaIN, WCT, SD-Turbo, StyleID, CUT, SaMST, SaMam, and Seedream 4.5. WEAVE uses four residual blocks (width 64), three velocity heads, and trains for 5 epochs with batch size 24 and AdamW.

Training and evaluation. WEAVE trains with AdamW (weight decay, peak LR $1{\times}10^{-4}$), gradient clipping at norm 1.0, and no EMA on a single RTX 3060 (12 GB). All methods are evaluated on every ordered source-target pair; learned baselines are selected by best validation CLIP-S on a 50-pair held-out split to avoid selection bias. CLIP-S is measured against the target-style pool and LPIPS against the ground-truth target (or source reconstruction under the IDT convention), with deterministic seeds throughout.



Main Results

IDT changes what counts as success. The identity row turns CLIP-S into a direction-aware metric: once included, low LPIPS alone is not enough, and both SaMST and SaMam stay below the IDT floor, suggesting their low LPIPS reflects source preservation.

WEAVE occupies the practical frontier. Among trained methods, WEAVE is the only row combining positive-IDT transfer with sub-0.30 LPIPS; CUT clears the floor only at higher cost and content distortion, while SaMST and SaMam preserve content but fall below the IDT floor. It thus achieves the strongest trained positive-IDT result on a single consumer GPU. The full multi-metric profile across all three benchmarks and both speed axes is summarized in Figure 6.

Empirical Efficiency. Our simplified wavelet architecture converges fast: empirically, WEAVE reaches its best results in 5 epochs (3.08 minutes on a single RTX 3060), avoiding the multi-hour optimization typical of dense flow matching. This speed comes from two properties of the decoupled design. First, by confining supervised transport to the high-frequency subbands, the effective parameter space that the

Method	Class	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer
		CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑		min	750 imgs
Identity	identity	0.693	0.000	1.000	0.419	0.663	0.000	1.000	0.416	0.731	0.000	1.000	0.433	0	free	0 s
SD-Turbo	diffusion	0.693	0.003	0.922	0.484	0.674	0.603	0.279	0.341	0.767	0.449	0.538	0.505	$0^{+1.1B}$	free	5 m
StyleAligned	diffusion	0.780	0.869	0.239	0.675	0.768	0.786	0.310	0.612	0.824	0.829	0.315	0.649	$0^{+1.1B}$	free	77 m
Z-STAR	diffusion	0.784	0.347	0.549	0.449	0.786	0.332	0.526	0.514	0.822	0.384	0.526	0.514	$0^{+1.1B}$	free	3 h
StyleShot	diffusion	0.787	0.765	0.377	0.563	0.774	0.713	0.335	0.552	0.787	0.795	0.380	0.484	$0^{+2.4B}$	free	5.1 h
CUT	GAN	0.714	0.374	0.795	0.471	0.754	0.494	0.702	0.539	0.710	0.620	0.795	0.441	7.0M	322.6	5 m
SaMST	Mamba	0.618	0.749	0.145	0.271	0.709	0.398	0.838	0.501	0.667	0.612	0.615	0.461	8.3M	39.5	10 m
SaMam	Mamba	0.582	$\underline{0.243}$	0.812	0.477	0.677	0.205	0.870	0.505	0.712	0.227	0.925	0.503	8.8M	436.0	17.6 m
Seedream 4.5	diffusion	0.720	0.477	0.739	0.486	0.752	0.227	0.785	0.517	0.742	0.486	0.725	0.504	–	API	–
Ours	**WEAVE**	0.7216	0.3769	0.7526	0.4277	**0.683**	**0.203**	0.878	0.510	**0.743**	**0.290**	0.816	0.480	**903K**$^{+84M}$	**3.08**	**83.7 s**

Table 1: Main comparison on D5-512, P2A-256, and R5-WikiArt. CLIP-S, DINO-C, and DINO-S are higher better; LPIPS is lower better. The D5 WEAVE entry is withheld pending same-checkpoint four-metric re-evaluation. Superscripts in Params denote frozen backbones, and "–" denotes withheld or unavailable values. Infer is single-RTX-3060 (12 GB) wall-clock for 750 pairs (per-image latency ×750).

optimizer must explore shrinks substantially—the LL subband, which carries the dominant signal energy, never enters the style-conditioned loss and therefore demands no learned displacement, only passive drift along the rectified-flow trajectory. Second, the endpoint-only statistical alignment removes the per-step gradient interactions that would otherwise couple the optimization of successive ODE steps, flattening the effective loss landscape into a set of largely independent sub-problems. While a rigorous convergence analysis remains beyond our scope, the ablation results (Section) are consistent with this picture: configurations that re-introduce frequency coupling (e.g., routing probability $p{=}0.5$, or per-step AdaIN injection) systematically degrade convergence speed, requiring more epochs to approach the same quality frontier.

Mechanistic diagnosis of the competing regimes. As the frequency-dominance analysis predicts, SaMam shows the style-suppression pattern—sub-IDT CLIP-S (0.5816) despite low LPIPS (0.2434)—while StyleID shows the predicted exponential decay (per-step injection drives $r_n(\alpha) = (1-\alpha)^n$, LPIPS 0.5523). WEAVE's endpoint-only WCT avoids this decay, reaching comparable positive-IDT transfer at much lower damage.

Large priors still set the style-strength ceiling. StyleID prioritizes raw CLIP-S, whereas WEAVE targets correct-direction transfer with far lower damage and cost: relative to Seedream 4.5 it cuts LPIPS by 39.8% with under one million parameters, and the entire system runs on a single consumer GPU.

Auxiliary metrics agree with the main reading. Figure 1(right) shows target-pooled ArtFID of 300.9 for WEAVE versus 311.5 for Seedream 4.5, placing both in the same realism range; on all-pairs evaluation WEAVE reaches 0.7213 CLIP-S / 0.2868 LPIPS, confirming the conclusion under the ablation protocol.

Mechanistic Ablations

Our subtractive audit shows that modules standard in prior work—optimal-transport routing, multi-step guidance, DINO correspondences—are not only unnecessary but often harmful here: The frequency-dominance phenomenon discussed in Section 2.3 indicates that any component coupling structure and texture in one latent basis is pulled toward favoring structure at the expense of style. We remove them, leaving a minimal design the optimizer can use for target-direction transport.

The $512{\times}512$ audit spans 43 configurations across seven theoretical axes (AdaIN $4\times$ retained in the CSV but omitted from the main text for space).

Most configurations cluster tightly around the full model, confirming the robustness of WEAVE's core design. The most prominent off-frontier point is the extrapolation 1 variant, whose content preservation collapses when high-frequency routing is removed, validating the role of endpoint subband alignment. A separate $10\times$ learning-rate run diverged to NaN and is excluded from the plot. The AdaIN $4\times$ variant was also omitted for readability because it caused severe structure distortion.

Axis-by-axis summary. The seven axes are LL weight $\lambda_{LL} \in \{0, 0.1, 0.3, 0.5, 1.0\}$ (0 removes low-frequency supervision), routing probability $p \in \{0.5, 0.8, 1.0\}$, endpoint scale vector, WCT on/off, RMSNorm vs. GroupNorm, residual blocks (2 vs. 4), and width (32 vs. 64). Across 43 configurations the full model is a clear Pareto optimum: no single-axis change improves both CLIP-S and LPIPS, and removing high-frequency routing ($p{=}0.5$) causes the largest CLIP-S drop (-0.032), while more blocks yield diminishing returns.

Why the cluster is tight. Most tested configurations cluster tightly, indicating moderate sensitivity to the scanned hyperparameters. We avoid interpreting this finite sweep as evidence of a globally flat or separable loss landscape.

Generalization and Robustness

Latent space vs. RGB. A matched $256{\times}256$ control shows the gain comes from wavelet transport in VAE latent space, not an extra loss: direct RGB flow reaches only 0.5842 CLIP-S / 0.4121 LPIPS, far worse than WEAVE, and the gap persists even at matched training time and parameters.

Few-shot style adaptation. Because WEAVE decouples content (LL) from style injection (LH/HL/HH), the high-frequency transport is not tied to a frozen vocabulary. Adapting on 8 held-out styles gives 0.7039 CLIP-S / 0.2555 LPIPS. Freezing the backbone and updating only the style-memory (30 images/style) already clears the IDT floor on all 8; fine-tuning the backbone adds only +0.008 CLIP-S at higher cost.

Random5-WikiArt generalization. To rule out cherry-picking, we train WEAVE from scratch on all 20 WikiArt families ($\geq$530 images each) and evaluate all 12,000 pairs, obtaining 0.7434 CLIP-S / 0.2904 LPIPS and clearing the Random5 identity floor of 0.7312 with LPIPS < 0.30. On the 750-pair 5-style subset, the baseline WEAVE reaches MUSIQ 44.24; applying semantic region SWD plus EOTA raises MUSIQ to 53.73 at 0.7182 CLIP-S / 0.3735 LPIPS, confirming the semantic-SWD mechanism generalizes beyond Distinct5. StyleID reaches the highest raw CLIP-S (0.8180) but at 0.5827 LPIPS; SD-Turbo and CUT lie between the high-style/high-distortion and low-style/low-distortion regimes, while SaMST stays below the identity floor (0.6669). The per-family breakdown shows consistent positive-IDT transfer everywhere, weakest on Pop Art and strongest on Ukiyo-e and Expressionism.

Wavelet basis. Daubechies-2 slightly raises CLIP-S but worsens LPIPS from 0.2868 to 0.3288. We retain Haar for its exact orthogonality and local support, which keeps both the theory and implementation simple while effectively avoiding blocking artifacts.

Resolution robustness. A 512-trained model evaluated on 256×256 inputs via inference-time downsampling (no retraining) drops only -0.018 CLIP-S and $+0.031$ LPIPS, showing the subband transport transfers to lower resolutions; upsampling a 256-trained model to 512 is far worse (-0.047 / $+0.069$), confirming high-frequency bands benefit from native-resolution training.

Discussion

Limitations. WEAVE is domain-conditional (a style family, not a reference painting). The Distinct5 benchmark is intentionally narrow (five hard, low-IDT styles as a no-op stress test), partially addressed by the Random5-WikiArt extension; the LL-protecting design makes palette-heavy targets like Minimalism harder than textured ones. The theory is local and mechanism-oriented; broader cross-domain evaluation would strengthen the empirical case.

Conclusion

Style transfer needs both the right success criterion and the right coordinate system. IDT calibration exposes when a method merely reconstructs the source, and WEAVE avoids this regime by separating low-frequency structure from high-frequency style motion and routing style through the same high-frequency path used at inference. The 903K-parameter backbone trains in 3.08 minutes on one RTX 3060. The D5 four-metric result is withheld until the same-checkpoint evaluation is complete; the current probes support band-aware modeling without claiming a complete content–style factorization. The central finding: real target-direction transfer does not require a large model once the transport geometry separates structure from style, and by running on a single consumer GPU WEAVE makes high-quality stylization accessible.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., "Impressionism") rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). The experiments section gives the training configuration, hyperparameters, and evaluation protocol. Training takes 3.08 minutes on a single RTX 3060 (12 GB). The 750-pair evaluation run needs 83.7 seconds for generation plus VAE decode and 97.3 seconds including CLIP/LPIPS metrics. The 903K trainable parameter backbone checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

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